Indraneel Deshpande

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Education

Indian Institute of Technology, Indore **2016 - 2020**
Bachelor of Technology in Metallurgy Engineering and Materials Science

Professional Experience

Associate, Neo Wealth and Asset Management, Mumbai **Jun 2025 - Present**

- Played a key role in scaling the structured products book from inception to **100 Cr AUM**, covering **principal-protected products** linked to equity indices and commodities such as Gold.
- Helped price and test structured products, including principal-protected notes constructed using **vanilla options** on equity indices and commodity underlyings.
- Developed a realistic end-to-end simulation incorporating all cost components (option premiums, hedging costs, funding, transaction costs) to assess **product feasibility** before launch.
- Automated **daily reporting** for the structured products book, tracking product performance metrics and real-time risk exposures across the portfolio.
- Collaborated with traders to build a **PnL attribution framework** decomposing option position performance into **Delta, Gamma, Theta, Vega, Vanna, Volga, and Residual PnL** components, enabling precise risk factor identification.
- Worked with traders to backtest strategies including **short volatility** and other derivatives-based approaches; improved strategy **Sharpe ratio** through dynamic hedging overlays and systematic **position sizing** to control tail risk and drawdowns.

Quantitative Strategist, Skylife Research, Mumbai **Oct 2024 - May 2025**

- Designed and implemented **Event-driven** and **Short volatility** trades based on statistical analysis of implied-realized volatility spread; used **Monte Carlo simulations** on straddles to generate robust PnL distribution estimates across multiple price paths.
- Conducted volatility research across equity indices and Nifty 50 stocks, identifying clustering patterns, seasonality, and price ranges; developed **OHLC-based volatility forecasts** for positional option trading.
- Built **fixed and floating strike volatility surfaces** and a Spot-Vol scenario grid to extract ATM dynamics, skew evolution, and vol-of-vol metrics for derivative pricing and trade timing.
- Developed and backtested **Momentum** and **Low Volatility** factor portfolios with transaction cost modeling; implemented risk controls to mitigate momentum crashes and improve Sharpe ratio.
- Conducted **network analysis** on equity markets using graph theory, measuring centrality metrics to improve portfolio diversification.

Quantitative Research Intern, Decibel Capital, New York (Remote) **Jul 2022 - Aug 2022**

- Implemented **Black-Scholes** pricing and calculated **implied volatility** via **Newton-Raphson** across a dataset of over a million Nifty option prices; attributed **Greeks** (Delta, Gamma, Vega, Theta) to PnL for straddle strategies.
- Analyzed futures data and implemented a **Time-to-expiry** function to assess the impact on **ATM straddles**; leveraged **PostgreSQL** and **TimescaleDB** for efficient data storage and retrieval.

Summer Intern, Zone Startups India, Bombay Stock Exchange **May 2019 - Jul 2019**

- Researched startups across FinTech, Energy, and Personal Finance for alignment with accelerator programs; compiled reports on CSR programs and government schemes such as NIDHI SSS.
- Facilitated strategic partnerships between clients and technology startups with expertise in machine learning.

Intern, Catalyst Partners, Mumbai **Dec 2018 - Jan 2019**

- Researched startup decks and funding histories using Venture Intelligence and Tracxn; facilitated investor-client connections to support the firm's strategic advisory services.

Projects

Derivatives and Volatility

- Developed an **Excel** model to replicate the payoff of an option using **Monte Carlo** simulation.
- Calculated **Greeks** and performed Delta, Theta, Gamma, and Residual PnL attribution to a call option using **VBA**.
- Simulated **Delta hedging** strategies and calculated cumulative PnL in Excel.
- Generated multiple stock paths with the same volatility using **Geometric Brownian Motion** and observed **Path dependency** by calculating Delta hedging PnL using Python.
- Wrote functions for different types of volatility, analyzed the volatility distribution, explored **volatility clustering**, and compared 30-day Nifty vol to India VIX.

Equity Portfolio Optimization

- Developed a framework for portfolio optimization using the Markowitz **Mean-Variance** approach with `numpy` and `cvxpy`, plotting Portfolio Return vs. Risk for different Risk aversion coefficients.
- Generated a **Resampled efficient frontier** through Monte Carlo simulation; calculated key risk metrics: **Sharpe ratio**, **VaR**, and **CVaR**.
- Enhanced portfolio optimization (**EPO**) for 15 stocks based on Pedersen's paper using shrinkage (w = 0.75), achieving Sharpe ratio of 1.67 over standard MVO; applied **Ledoit-Wolf** and **Ridge** regularization to improve Sample Covariance matrix conditioning.
- Constructed **Equal Risk Contribution** portfolios via Risk Budgeting; applied **PCA** to build Eigenportfolios and created a **Barra**-aligned Factor exposure matrix with factor covariance and idiosyncratic variance.

Skills

- **Quantitative Analysis**: Portfolio optimization, Factor modeling, Risk measurement, Monte Carlo simulation, Statistical Analysis, Derivatives, Volatility, Structured Products.
- **Programming Languages**: Python (`numpy`, `pandas`, `matplotlib`, `cvxpy`, `networkx`), SQL (PostgreSQL), VBA, Bash.
- **Tools & Technologies**: Linux, Jupyter Notebook, Excel, Solver.
- **Methodologies**: Mean-Variance optimization, Covariance matrix shrinkage, Risk budgeting, Multi-factor modeling, Delta Hedging, PnL Attribution, Momentum Strategy Design.
- **Collaboration**: Effective teamwork with traders, risk, portfolio, and business managers to implement quantitative models.

Publications

- **Chemistry Journal**: Substituents Effects of Organic Linkers on Rotational Energy Barriers in Metal-Organic Frameworks

Relevant Courses

- Linear Algebra (IIT Indore)
- Probabilistic and Statistical Methods in Civil Engineering (IIT Indore)
- Computational Methods in Pricing and Model Calibration (Columbia University, Coursera)
- Advanced Topics in Derivative Pricing (Columbia University, Coursera)
- Optimization Methods in Asset Management (Columbia University, Coursera)

- Neural Networks and Deep Learning
- Improving Deep Neural Networks: Hyperparameter tuning, Regularization and Optimization
- Structuring Machine Learning Projects